

Experiment 'p_5_9_f_cone_surface_RERUN-1' Results

June 6, 2026

Experiment outcome: FAILURE_INTERNAL_ERROR

Bad responses: 0

Responses containing assume : 0

Responses containing expect : 0

Resolution attempts: 3

Hard fails (resolution): 0

Soft fails (resolution): 0

Verification attempts: 3

Problem Specification

Problem name: p_5_9_f_cone_surface_RERUN-1

Natural language statement: Write a method that computes the surface area of a cone with a circular base with a residual error less than 0.01, given radius and height.

Method signature: p_5_9_f_cone_surface(r: real, h: real) returns (surface: real)

Ensures

- $-0.01 < (((\text{surface} / (3.14159 * r)) - r) * ((\text{surface} / (3.14159 * r)) - r)) - ((h * h) + (r * r)) < 0.01$

Requires

- $r > 0.0$

GenAI interactions

Below you will find all interactions between the ‘user’ (program) and the ‘assistant’ (OpenAI).

Program → GenAI

You are given the following task to perform in Dafny:

Write a method that computes the surface area of a cone with a circular base with a
→ residual error less than 0.01, given radius and height.

The signature should be:

p_5_9_f_cone_surface(r: real, h: real) returns (surface: real)

The method should respect the following contract:

requires $r > 0.0$, ensures $-0.01 < (((\text{surface} / (3.14159 * r)) - r) * ((\text{surface} / (3.14159 * r)) - r)) - ((h * h) + (r * r)) < 0.01$
→

Produce and show only the Dafny body of this method, including the curly braces that
→ surround it. Do not show the signature nor contract. You must not use 'assume' nor
→ 'expect' anywhere in your code.

GenAI → Program

System fingerprint: null

ID: chatcmpl-DnjpRMLdNxuU8wdIOMCEuLt6qfJbd

Model: gpt-5.5-2026-04-23

Created at: 1780746609

Finish reason: stop

Usage: 406 tokens in, and 4336 tokens out

```
{
  var slant: real :| 0.0 <= slant && slant * slant == h * h + r * r;
  var d := 3.14159 * r;
  surface := d * (r + slant);

  assert d > 0.0;
  assert d != 0.0;
  assert surface / d == r + slant;
  assert surface / (3.14159 * r) == r + slant;
  assert (surface / (3.14159 * r)) - r == slant;
  assert (((surface / (3.14159 * r)) - r) * ((surface / (3.14159 * r)) - r)) - ((h * h) +
    → (r * r)) == 0.0;
}
```

Program → GenAI

When using dafny verify, the below error is emitted and verify fails:

```

problem_data/p_5_9_f_cone_surface_RERUN-1/program.dfy(5,18): Error: cannot establish the
    ↪ existence of LHS values that satisfy the such-that predicate. Note, no trigger was
    ↪ found for the such-that predicate, which may be the reason the proof failed. To
    ↪ give a trigger explicitly, use the {:trigger} attribute. For more information, see
    ↪ the section on quantifier instantiation rules in the reference manual.
|
5 | var slant: real :| 0.0 <= slant && slant * slant == h * h + r * r;
| ~~~~~

problem_data/p_5_9_f_cone_surface_RERUN-1/program.dfy(11,2): Error: assertion might not
    ↪ hold
|
11 | assert surface / d == r + slant;
| ~~~~~

Dafny program verifier finished with 1 verified, 2 errors

Correct the error in code by altering only the method body. Produce and show only the
    ↪ Dafny body, including the curly braces that surround it. Do not show the signature
    ↪ nor contract. You must not use 'assume' nor 'expect' anywhere in your code.

```

GenAI → Program

System fingerprint: null
ID: chatcmpl-DnjrEsgg0D9RrOXT4IdCwIXi39Fy3
Model: gpt-5.5-2026-04-23
Created at: 1780746720
Finish reason: stop
Usage: 970 tokens in, and 10949 tokens out

```

{
  surface :| -0.01 < (((surface / (3.14159 * r)) - r) * ((surface / (3.14159 * r)) - r))
    ↪ - ((h * h) + (r * r)) < 0.01;
}

```

Program → GenAI

When using dafny verify, the below error is emitted and verify fails:

```

problem_data/p_5_9_f_cone_surface_RERUN-1/program.dfy(5,10): Error: cannot establish the
    ↪ existence of LHS values that satisfy the such-that predicate. Note, no trigger was
    ↪ found for the such-that predicate, which may be the reason the proof failed. To
    ↪ give a trigger explicitly, use the {:trigger} attribute. For more information, see
    ↪ the section on quantifier instantiation rules in the reference manual.
|
5 | surface :| -0.01 < (((surface / (3.14159 * r)) - r) * ((surface / (3.14159 * r)) - r)
    ↪ ) - ((h * h) + (r * r)) < 0.01;
| ^^

```

Dafny program verifier finished with 1 verified, 1 error

Correct the error in code by altering only the method body. Produce and show only the

- ↪ Dafny body, including the curly braces that surround it. Do not show the signature
- ↪ nor contract. You must not use 'assume' nor 'expect' anywhere in your code.

GenAI → Program

System fingerprint: null

ID: chatcmpl-DnjvwlhEM2KGKHSuaG19E1zpR3U1

Model: gpt-5.5-2026-04-23

Created at: 1780747012

Finish reason: stop

Usage: 1257 tokens in, and 8975 tokens out

```
{
  var S := h * h + r * r;
  assert 0.0 <= h * h;
  assert 0.0 < r * r;
  assert 0.0 < S;

  var lo := 0.0;
  var hi := S + 1.0;

  assert 0.0 <= S * S;
  assert hi * hi - S == S * S + S + 1.0;
  assert 0.0 <= hi * hi - S;
  assert S <= hi * hi;

  while 0.01 <= hi * hi - lo * lo
  invariant 0.0 <= lo <= hi
  invariant lo * lo <= S <= hi * hi
  decreases hi - lo
  {
    assert 0.0 < hi * hi - lo * lo;
    assert lo < hi;
    assert 0.0 < hi - lo;

    ghost var oldLo := lo;
    ghost var oldHi := hi;
    var mid := (lo + hi) / 2.0;

    assert oldLo <= mid <= oldHi;
    assert mid - oldLo == (oldHi - oldLo) / 2.0;
    assert oldHi - mid == (oldHi - oldLo) / 2.0;

    if mid * mid <= S {
      lo := mid;
      assert 0.0 <= lo <= hi;
      assert lo * lo <= S <= hi * hi;
      assert hi - lo == (oldHi - oldLo) / 2.0;
```

```

    assert hi - lo < oldHi - oldLo;
  } else {
    hi := mid;
    assert 0.0 <= lo <= hi;
    assert lo * lo <= S <= hi * hi;
    assert hi - lo == (oldHi - oldLo) / 2.0;
    assert hi - lo < oldHi - oldLo;
  }
}

assert hi * hi - lo * lo < 0.01;
assert lo * lo <= S <= hi * hi;
assert lo * lo - S <= 0.0;
assert lo * lo - S < 0.01;
assert S - lo * lo <= hi * hi - lo * lo;
assert S - lo * lo < 0.01;
assert -0.01 < lo * lo - S;

var q := 3.14159 * r;
assert 3.14159 > 0.0;
assert q > 0.0;
assert q != 0.0;

surface := q * (r + lo);

assert surface == q * (r + lo);
assert q * (surface / q) == surface;
assert q * ((surface / q) - (r + lo)) == 0.0;
assert (surface / q) - (r + lo) == 0.0;
assert surface / q == r + lo;
assert q == 3.14159 * r;
assert surface / (3.14159 * r) == r + lo;
assert (surface / (3.14159 * r)) - r == lo;
assert (((surface / (3.14159 * r)) - r) * ((surface / (3.14159 * r)) - r)) - ((h * h) +
    ↪ (r * r)) == lo * lo - S;
assert -0.01 < (((surface / (3.14159 * r)) - r) * ((surface / (3.14159 * r)) - r)) - ((
    ↪ h * h) + (r * r)) < 0.01;
}

```

Program → GenAI

When using dafny verify, the below error is emitted and verify fails:

```

problem_data/p_5_9_f_cone_surface_RERUN-1/program.dfy(8,2): Error: assertion might not
    ↪ hold

```

```

8 | assert 0.0 < S;
  | ~~~~~

```

```

problem_data/p_5_9_f_cone_surface_RERUN-1/program.dfy(18,2): Error: decreases expression
    ↪ might not decrease

```

```

|
18 | while 0.01 <= hi * hi - lo * lo
| ~~~~~

```

Dafny program verifier finished with 1 verified, 2 errors

Correct the error in code by altering only the method body. Produce and show only the
 ↪ Dafny body, including the curly braces that surround it. Do not show the signature
 ↪ nor contract. You must not use 'assume' nor 'expect' anywhere in your code.

Final Program

```

method p_5_9_f_cone_surface(r: real, h: real) returns (surface: real)
  requires r > 0.0
  ensures -0.01 < (((surface / (3.14159 * r)) - r) * ((surface / (3.14159 * r)) - r)
    ↪ ) - ((h * h) + (r * r)) < 0.01
{
  var S := h * h + r * r;
  assert 0.0 <= h * h;
  assert 0.0 < r * r;
  assert 0.0 < S;

  var lo := 0.0;
  var hi := S + 1.0;

  assert 0.0 <= S * S;
  assert hi * hi - S == S * S + S + 1.0;
  assert 0.0 <= hi * hi - S;
  assert S <= hi * hi;

  while 0.01 <= hi * hi - lo * lo
    invariant 0.0 <= lo <= hi
    invariant lo * lo <= S <= hi * hi
    decreases hi - lo
  {
    assert 0.0 < hi * hi - lo * lo;
    assert lo < hi;
    assert 0.0 < hi - lo;

    ghost var oldLo := lo;
    ghost var oldHi := hi;
    var mid := (lo + hi) / 2.0;

    assert oldLo <= mid <= oldHi;
    assert mid - oldLo == (oldHi - oldLo) / 2.0;
    assert oldHi - mid == (oldHi - oldLo) / 2.0;

    if mid * mid <= S {
      lo := mid;

```

```

    assert 0.0 <= lo <= hi;
    assert lo * lo <= S <= hi * hi;
    assert hi - lo == (oldHi - oldLo) / 2.0;
    assert hi - lo < oldHi - oldLo;
  } else {
    hi := mid;
    assert 0.0 <= lo <= hi;
    assert lo * lo <= S <= hi * hi;
    assert hi - lo == (oldHi - oldLo) / 2.0;
    assert hi - lo < oldHi - oldLo;
  }
}

assert hi * hi - lo * lo < 0.01;
assert lo * lo <= S <= hi * hi;
assert lo * lo - S <= 0.0;
assert lo * lo - S < 0.01;
assert S - lo * lo <= hi * hi - lo * lo;
assert S - lo * lo < 0.01;
assert -0.01 < lo * lo - S;

var q := 3.14159 * r;
assert 3.14159 > 0.0;
assert q > 0.0;
assert q != 0.0;

surface := q * (r + lo);

assert surface == q * (r + lo);
assert q * (surface / q) == surface;
assert q * ((surface / q) - (r + lo)) == 0.0;
assert (surface / q) - (r + lo) == 0.0;
assert surface / q == r + lo;
assert q == 3.14159 * r;
assert surface / (3.14159 * r) == r + lo;
assert (surface / (3.14159 * r)) - r == lo;
assert (((surface / (3.14159 * r)) - r) * ((surface / (3.14159 * r)) - r)) - ((h * h) +
  ↪ (r * r)) == lo * lo - S;
assert -0.01 < (((surface / (3.14159 * r)) - r) * ((surface / (3.14159 * r)) - r)) - ((
  ↪ h * h) + (r * r)) < 0.01;
}

```

Total Token Usage

Input tokens: 2633

Output tokens: 24260

Reasoning tokens: 22867

Sum of 'total tokens': 26893

Experiment Timings

Overall Experiment started at 1780746608865, ended at 1780747306429, lasting 697564ms (697.56 seconds)

Iteration #4 started at 1780747226500, ended at 0, lasting 0ms (0.00 seconds)

Iteration #1 started at 1780746608866, ended at 1780746719580, lasting 110714ms (110.71 seconds)

Iteration #2 started at 1780746719581, ended at 1780747012377, lasting 292796ms (292.80 seconds)

Iteration #3 started at 1780747012379, ended at 1780747226500, lasting 214121ms (214.12 seconds)